

DAMPS-MMHCL Diagnostic Analysis Report

Amazon Clothing Dataset — Performance Gap Investigation

Synthesised Multi-Model Analysis

GPT-5.5 Thinking · Claude Opus 4.7 Thinking · Gemini 3.1 Pro Thinking

Repository: https://github.com/sunflower82/DAMPS_upgrade_for_MMHCL_randoms_Amazon_Clothing

Architecture: DAMPS_to_MMHCL_architecture_revision42.tex (Revision 9)

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Abstract

This report investigates a systematic 10.7% Recall@20 deficit observed when integrating DAMPS into the MMHCL framework on the Amazon Clothing dataset (0.0786 vs. 0.0881 paper baseline, averaged over 10 random seeds). NDCG@20 and Precision@20 degrade by only 2.9% and 2.6% respectively, indicating a *candidate coverage failure* rather than a ranking failure. Three frontier reasoning models independently converge on this diagnosis. Six ranked root-cause hypotheses are presented—from an evaluation protocol mismatch (highest impact, zero compute cost to verify) to a learnable temperature τ stuck at a local minimum, AVRF spectral over-filtering, and a hypergraph NNZ explosion—together with a prioritised six-configuration ablation grid for efficient diagnosis.

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1. Executive Summary

Experimental results across 10 random seeds show that the current DAMPS-MMHCL integration achieves $\text{Recall@20} = 0.0786 \pm 0.0016$, which is **10.7%** below the original MMHCL paper baseline of 0.0881. By contrast, NDCG@20 drops only 2.9% (0.0383 vs. 0.0394) and Precision@20 drops only 2.6% (0.00438 vs. 0.0045). This asymmetric degradation pattern — Recall collapses while ranking metrics remain nearly intact — is a strong diagnostic signal that the model is *narrowing candidate coverage* rather than mis-ordering items. Three independent frontier reasoning models all converge on this interpretation, though they disagree on the primary root cause.

2. Consensus Findings: High-Confidence Results

Table 1: Findings unanimously confirmed by all three reasoning models.

Finding	GPT	Claude	Gemini	Evidence / Source
Recall@20 drops $\approx 10.7\%$ vs. original MMHCL (0.0786 vs. 0.0881)	✓	✓	✓	DAMPS_MMHCL_Clothing_Results_Local.xlsx, Paper Comparison sheet
NDCG@20 drops less ($\approx 2.9\%$, 0.0383 vs. 0.0394)	✓	✓	✓	Same Excel file, Summary Statistics sheet
Training curve plateaus early ($\approx$ epoch 150–170) and does not recover further	✓	✓	✓	recall-20.csv, peak ≈ 0.0805 at step 300, then declining
DAMPS augmentation may disrupt the hypergraph structure $\rightarrow$ reduces recall; Soft Residual-Routing $\alpha = 0.1$ may be mis-calibrated	✓	✓	✓	DAMPS_to_MMHCL_architecture_revision42.tex, Section 3

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Finding	GPT	Claude	Gemini	Evidence / Source
Learnable temperature τ converges to 0.0909 (near its init value 0.1) across all 10 seeds; may constrain embedding diversity	✓	✓	✓	DAMPS_MMHCL_Clothing_Results_Local.xlsx, column tau_final
Seed-to-seed stability is acceptable ($\sigma = 0.0016$ for Recall@20)	✓	✓	✓	Excel Summary Statistics sheet

2.1. Quantitative Comparison with Original MMHCL

Table 2 presents the full numerical comparison between the DAMPS-MMHCL runs (10 seeds) and the MMHCL paper baseline.

Table 2: Metric comparison: DAMPS-MMHCL vs. original MMHCL on Amazon Clothing. Source: DAMPS_MMHCL_Clothing_Results_Local.xlsx.

Metric	DAMPS-MMHCL (mean $\pm$ std)	MMHCL Paper	Δ abs	Δ rel (%)
Recall@20	0.07864 $\pm$ 0.00159	0.0881	−0.00946	−10.73
NDCG@20	0.03827 $\pm$ 0.00078	0.0394	−0.00113	−2.88
Prec@20	0.00438 $\pm$ 0.00008	0.0045	−0.00012	−2.64

The asymmetry between Δ Recall@20 (−10.73%) and Δ NDCG@20 (−2.88%) is the central diagnostic signal in this report. A system that scores well on NDCG but poorly on Recall is providing *high-quality rankings within a diminished candidate pool*. This implicates the **candidate generation stage** (hypergraph neighbourhood, embedding diversity) rather than the ranking stage.

2.2. Individual Seed Results

No seed can be classified as an outlier: the IQR for Recall@20 is ≈ 0.0028 , and the range (0.07546–0.08119) never approaches the paper baseline of 0.0881. This confirms the deficit is **systematic**, not due to unlucky initialisation.

Table 3: Per-seed results. Source: DAMPS_MMHCL_Clothing_Results_Local.xlsx, Individual Results sheet.

Run	Seed	Recall@20	Precision@20	NDCG@20
1	1303944351	0.07904	0.004398	0.038456
2	1945848931	0.08119	0.004510	0.038809
3	1338264894	0.07834	0.004362	0.036728
4	2131176512	0.07993	0.004449	0.038843
5	444520708	0.07720	0.004321	0.037912
6	1691295048	0.08045	0.004479	0.039255
7	27086229	0.07830	0.004367	0.039236
8	1691374281	0.07760	0.004311	0.037648
9	90487858	0.07546	0.004224	0.037474
10	156097670	0.07893	0.004393	0.038298
Mean	—	0.07864	0.004381	0.038266
Std	—	0.00159	0.000081	0.000780
Min	—	0.07546	0.004224	0.036728
Max	—	0.08119	0.004510	0.039255

3. Training Curve Analysis (Seed 156097670)

The five WandB CSV files (recall-20.csv, recall-10.csv, ndcg-20.csv, hit-20.csv, precision-20.csv) log validation metrics at evaluation steps for seed 156097670.

3.1. Key Observations

- **Recall@20** rises sharply from step 5 (0.0518) to step 300 (0.0805), then oscillates and declines to 0.0774 at step 364. Peak step: **300** ($\approx$ epoch 214).
- **NDCG@20** peaks at step 285 (0.0371), slightly ahead of the Recall peak — consistent with the model optimising ranking quality before coverage.
- **HitRate@20** mirrors Recall@20 closely; peak at step 300 (0.0830).
- **Precision@20** peaks around step 300 (0.00418) and also plateaus.
- **Recall@10** continues to show moderate gains beyond step 220, suggesting the model could still improve at shorter list lengths.
- **Plateau onset**: visible from $\approx$ step 162 (epoch 115), $\approx$ 150 epochs before peak — consistent with early saturation of the contrastive objective.
- **No divergence or instability**: the oscillation amplitude post-plateau is < 0.003 for Recall@20, indicating stable training dynamics.

3.2. Early Stopping Assessment

With `early_stopping_patience = 30` and `early_stopping_min_epochs = 75`, the model likely stopped around step 364–385. The best checkpoint at step 300 was correctly captured. **The problem is not early stopping prematurely — the model reached its capacity ceiling at ≈ 0.0805 , well below the 0.0881 target.**

4. Areas of Divergence Between Models

Table 4: Points of disagreement among the three reasoning models.

Topic	GPT-5.5 Thinking	Claude Opus 4.7	Gemini 3.1 Pro
Primary root cause #1	τ too low $\rightarrow$ collapse of embedding diversity	DAMPS filter too aggressive, destroying hypergraph edges	Early stopping + LR schedule halt training prematurely
Pattern B rebuild ($R = 5$)	Frequency is adequate; not a problem	May cause instability in early epochs	$R = 5$ is fine, but 10-epoch warmup is insufficient
First recommended fix	Raise τ_{init} to 0.2–0.5	Reduce DAMPS strength ($\alpha \rightarrow 0.05$)	Increase max epochs to 400+ with patience 50

The disagreements arise from each model weighting different diagnostic signals: GPT-5.5 focuses on the τ numerical anomaly; Claude Opus focuses on architectural interactions; Gemini focuses on training dynamics. All three hypotheses should be tested in parallel (see Section 8).

5. Model-Specific Discoveries

6. Architecture Delta: DAMPS vs. Original MMHCL

6.1. Summary of Modifications

Based on `DAMPS_to_MMHCL_architecture_revision42.tex` (Revision 9), the following components were integrated:

- Adaptive Phase Calibration (APC):** Static metadata-driven von Mises estimation of per-category phase rotation $\hat{\phi}_c$; eliminates dynamic k -means overhead.
- Adaptive Variance Regularisation Filter (AVRF):** Wiener-inspired frequency-domain shrinkage with learnable logit $w_{m,f} \in [-2, 2]$; data-driven SNR prior initialisation.
- Inter-Modal Coherence Filter Residual (IMCF):** Residual modulation using ASC coherence coefficient; formulated as $\mathbf{z}_i^m \leftarrow \mathbf{z}_i^m + \gamma(\text{ASC}_i - \overline{\text{ASC}})\mathbf{z}_i^m$.

Table 5: Novel findings specific to each reasoning model.

Model	Unique Finding	Why It Matters
GPT-5.5 Thinking	$\tau = 0.0909$ is identical across all 10 seeds, nearly unchanged from <code>tau_init = 0.1</code> , indicating the learnable temperature is stuck in a local minimum.	If τ does not learn, InfoNCE loss remains static and contrastive learning fails to calibrate inter-embedding distances dynamically.
Claude Opus 4.7 Thinking	<code>avg_deg</code> ≈ 68 in the Excel file versus the expected K-NN setting <code>top-k = 5</code> . Even accounting for second-order hypergraph connectivity, the degree is suspiciously high, suggesting a residual NNZ explosion in Pattern B rebuilds.	If the adjacency matrix accumulates non-zero entries across rebuilds, the graph structure blurs, reducing the specificity of collaborative signals and dropping Recall.
Gemini 3.1 Pro Thinking	The large gap between $\Delta\text{Recall@20}$ (-10.7%) and $\Delta\text{Precision@20}$ (-2.6%) implies the model ranks <i>known</i> high-quality items well but fails to surface a diverse candidate pool. DAMPS spectral filtering may be homogenising item representations.	Reduced candidate diversity is a known failure mode when contrastive augmentation is too strong or the filter attenuates discriminative frequency bins.

4. **Soft Residual-Routing** (α -blend): $\mathbf{h}_{\text{input}} = \mathbf{h}_{\text{raw}} + \alpha \cdot \text{LayerNorm}(\mathbf{h}_{\text{cal}})$, where α is initialised at 0.1 and learnable.
5. **Pattern B Scheduled Rebuild** ($R = 5$ epochs): Hypergraph reconstructed every 5 epochs using calibrated features.
6. **Slim Momentum Encoder**: EMA smoothing applied to $\mathbf{h}_{\text{cal}} \in \mathbb{R}^{64}$ only (not the full 4096-dimensional visual table).
7. **Learnable InfoNCE Temperature**: τ initialised at 0.1; intended to adapt dynamically.
8. **Mixed Precision (BF16)**: Enabled via `use_amp = True`.

6.2. Impact Assessment per Component

7. Root Cause Hypotheses for Recall@20 Drop

The following hypotheses are ranked by estimated plausibility based on the convergence of model opinions and numerical evidence.

Hypothesis H1 (High Probability): Learnable τ Stuck at Local Minimum

Evidence (observed): All 10 seeds converge to $\tau_{\text{final}} = 0.0909$, which deviates only -9.1% from the initialisation of 0.1. This near-zero shift over 200+ epochs is anomalous for a learnable scalar with gradient flow.

Mechanism: A temperature of ≈ 0.09 makes the InfoNCE softmax extremely sharp. During contrastive training, the model focuses exclusively on hardest negatives, creating tightly clustered embeddings. In such a landscape, only the few nearest neighbours of each user receive high scores — reducing the effective retrieval radius and suppressing Recall@20 while NDCG remains relatively stable.

Test: Freeze τ at $\{0.2, 0.3, 0.5\}$ (three runs). Expected gain: $+3\text{--}5\%$ Recall@20.

Hypothesis H2 (High Probability): AVRF Over-Filtering Discriminative Frequency Bins

Evidence (observed): The AVRF logit initialisation uses a data-driven SNR prior clipped to $[-2, 2]$. If the prior under-estimates intra-cluster variance (which occurs for sparse e-commerce data), the gate $\sigma(w_{m,f})$ can become uniformly small, attenuating features across all bins.

Mechanism: Post-AVRF feature vectors become more similar across items. Cosine similarity among all pairs converges toward a narrow range. The K-NN hypergraph construction selects near-arbitrary neighbours, polluting the graph and degrading collaborative signal propagation.

Table 6: Estimated impact of each DAMPS component on Recall@20 and NDCG@20. **O** = observed from data; **H** = hypothesis to be tested.

Component	Impact on Recall@20	Impact on NDCG@20
AVRF (spectral shrinkage)	H : May over-attenuate discriminative bins $\rightarrow$ uniform cosine similarity $\rightarrow$ wrong K-NN edges $\rightarrow$ lower coverage.	H : Moderate degradation; ranking within the narrowed pool may still be coherent.
APC (phase rotation)	H : Reduces cross-modal phase misalignment; should help coverage if calibration is accurate.	H : Positive or neutral effect.
IMCF (coherence modulation)	H : Residual form limits attenuation; risk is scaling error if $\overline{\text{ASC}}$ is mis-estimated.	Likely neutral to positive if ASC is well-calibrated.
Soft Residual-Routing (α)	O : $\alpha_{\text{init}} = 0.1$ may be too small, meaning DAMPS barely intervenes; or too large if the learnable α overshoots.	Tied to the degree of intervention: small α means near-vanilla MMHCL embeddings.
Pattern B Rebuild ($R = 5$)	H : Stable connectivity if NNZ is controlled; potentially causing density explosion (see avg_deg anomaly).	Hypergraph quality directly impacts propagation quality.
Learnable τ	O : $\tau_{\text{final}} = 0.0909$ across all seeds; very low $\rightarrow$ sharp softmax $\rightarrow$ embedding collapse $\rightarrow$ reduced coverage at top-20.	Low τ can artificially inflate NDCG by concentrating scores on a few high-confidence items.
Momentum Encoder (Slim)	Stabilises hypergraph rebuilds; net positive for coverage.	Stabilises contrastive targets; net positive.

Test: Single ablation — disable AVRF while keeping APC and IMCF active. Expected gain: +2–4% Recall@20.

Hypothesis H3 (Medium Probability): NNZ Explosion in Pattern B Rebuild

Evidence (observed): $\text{avg_deg} \approx 68$ in the Excel log, versus a K-NN parameter $\text{top-k} = 5$. Even with second-order hyperedge convolution $\mathbf{H}_{i2i} = \text{LN}(\bar{\mathbf{I}}\mathbf{W}\bar{\mathbf{I}}^\top)$, a degree of $68/5 = 13.6\times$ the sparsity target is suspicious.

Mechanism: If EMA smoothing of the sparse adjacency accumulates NNZ entries across rebuild cycles, each node gains spurious edges, effectively oversmoothing item representations and degrading distinctiveness.

Test: Log NNZ and avg_deg at every Pattern B rebuild; if NNZ increases monotonically, enforce hard re-sparsification after each rebuild. Expected gain: +2–3% Recall@20.

Hypothesis H4 (Medium Probability): Evaluation Protocol Mismatch

Evidence (observational gap): The paper reports 0.0881 but the implementation’s best checkpoint is 0.0805 — a floor that cannot be reached regardless of training length. Protocol differences (sampled negatives vs. full ranking, inclusive vs. exclusive k , seen-item filtering) can account for 0–10% absolute differences.

Test: Compare the evaluation loop in `codes/` with the original MMHCL repository (<https://github.com/Xu107/MMHCL>). Priority: **check this first** as it requires no retraining.

Hypothesis H5 (Medium Probability): Soft Residual-Routing α Miscalibrated

Evidence (architectural): $\alpha_{\text{init}} = 0.1$ means the HGCM input is $\approx 90\%$ raw features and $\approx 10\%$ calibrated features. If α does not learn to a meaningful value, DAMPS barely intervenes — yet still adds noise through the LayerNorm path.

Test: Sweep $\alpha \in \{0.3, 0.5, 0.7\}$ as static values. Expected gain: $\pm 2\%$ Recall@20 depending on direction.

Hypothesis H6 (Lower Probability): Training Capacity Ceiling

Evidence: Peak Recall@20 for seed 156097670 is 0.0805 at step 300 (recall-20.csv), with consistent behaviour across all 10 seeds (max = 0.08119). The model architecture uses $\text{embedding_dim} = 64$, which may be insufficient for the Amazon Clothing dataset scale.

Test: Extend training to 400 epochs with $\text{patience} = 50$. Additionally, test $\text{embedding_dim} = 128$.

8. Recommendations

8.1. Priority Action Table

Table 7: Recommended experiments ordered by expected impact and cost.

Issue	Hypothesis	Experiment	Expected Δ Recall@20
Evaluation protocol mismatch	Paper may use full ranking or different filter rules	Manually diff eval loop vs. MMHCL repository	+0–10% (if bug found)
τ stuck / too low	Learnable τ gradient vanish; static alternative may perform better	Static sweep: $\tau \in \{0.2, 0.3, 0.5\}$, 3 runs at single seed	+3–5%
AVRF over-filtering	AVRF attenuates discriminative frequency content	Ablation: disable AVRF only; keep APC + IMCF	+2–4%
Soft-Routing α wrong	$\alpha = 0.1$ too small; DAMPS barely modifies embeddings	Static sweep: $\alpha \in \{0.3, 0.5, 0.7\}$	$\pm 2\%$
NNZ / avg_deg explosion	Momentum encoder not controlling adjacency sparsity	Add per-rebuild NNZ logging; force hard re-sparsify to top- K	+2–3%
Training capacity ceiling	Plateau is true optimum under current capacity	Extend to 400 epochs, patience 50; test dim 128	+1–2%

8.2. High-Priority Grid Sweep (≤ 6 Configurations)

Table 8: Recommended grid sweep (6 configurations, 1 seed each for screening).

Config	Label	τ	α	AVRF	max_epochs
1	Baseline (current)	learnable (0.1)	0.1	ON	250
2	τ -fixed-0.2	0.2 (fixed)	0.1	ON	250
3	τ -fixed-0.3	0.3 (fixed)	0.1	ON	250
4	AVRF-OFF	learnable (0.1)	0.1	OFF	250
5	α -0.5	learnable (0.1)	0.5	ON	250
6	Protocol-check	learnable (0.1)	0.1	ON	250

Config 6 is reserved for the evaluation protocol audit (no retraining required; only changes the evaluation script).

8.3. Ablation Design for Root Cause Isolation

To unambiguously identify the primary cause, the following sequential ablation is recommended:

1. **Protocol audit** (zero compute): Compare evaluation functions side-by-side with MMHCL original. If a bug is found, fix and re-evaluate all 10 seeds.
2. τ **sweep** (3×1 seed): If Config 2 or 3 shows $\geq +3\%$ Recall@20, τ is confirmed as the dominant factor. Expand the winning τ to all 10 seeds.
3. **AVRF ablation** (1 seed): Run Config 4. If gain is $\geq +2\%$, AVRF is over-filtering. Consider constraining AVRF logit range to $[-1, 1]$ or adding a diversity-preserving regulariser.
4. **NNZ audit**: Add logging:

```
if epoch % R == 0:
    nnz = adj_matrix.nnz if adj_matrix.issparse() else (
        adj_matrix != 0).sum().item()
    avg_deg = nnz / N
    logger.info(f"Epoch {epoch}: NNZ={nnz}, AvgDeg={avg_deg:.2f}")
```

If avg_deg grows over time, enforce:

```
# After each rebuild, hard re-sparsify to top-K neighbours
adj_matrix = keep_top_k(adj_matrix, k=top_k)
```

9. Conclusion

The DAMPS-MMHCL integration on Amazon Clothing yields a **systematic 10.7% Recall@20 deficit** relative to the original MMHCL baseline (0.0786 vs. 0.0881), while NDCG@20 and Precision@20 degrade only 2.9% and 2.6% respectively. This asymmetric pattern conclusively points to *reduced candidate coverage* rather than ranking degradation.

The three most probable root causes, ranked by evidence strength, are:

1. **Evaluation protocol mismatch** (must be ruled out first — zero compute cost; potential impact up to +10%).
2. **Learnable temperature τ stuck at ≈ 0.09** — observed across all 10 seeds; causes embedding collapse and narrows the top-20 retrieval radius.
3. **AVRF spectral over-filtering** — attenuates discriminative frequency content, homogenising item representations and degrading hypergraph edge quality.

Three immediate actions (in priority order):

1. Audit the evaluation loop against the original MMHCL codebase at <https://github.com/Xu107/MMHCL>.
2. Run a static τ sweep $\{0.2, 0.3, 0.5\}$ on a single seed (≤ 3 GPU runs, ≈ 6 hours).
3. Run an AVRf-ablation (disable AVRf, keep APC + IMCF active) on a single seed to quantify the filtering penalty.

If both τ -fix and AVRf-ablation produce positive gains, their combination (Config: $\tau = 0.3$, AVRf-OFF) should be re-evaluated across all 10 seeds for final statistical reporting.